Frank Yingjie Huo

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Research interests

Exactly solvable models of collective dynamics in complex systems: many-body theory of Transformer-based AI, from a single attention head to a whole conversation; heterogeneous assembly and fragmentation dynamics on complex networks.

Education

2021 – 2027 **The George Washington University, Washington, DC**
PhD in Physics, expected May 2027
Advisor: Prof. Neil F. Johnson
Columbian Distinguished Fellow, Columbian College of Arts and Sciences

2017 – 2021 **Pembroke College, University of Cambridge, UK**
MMath (Part III of the Mathematical Tripos), 2021
BA in Physics (Natural Sciences Tripos), 2021; MA (Cantab), 2025
Foundation Scholar; Foundress Prize

Awards and honors

2026 SIAM Student Travel Award, SIAM Annual Meeting (AN26)
2026 Scholar, Edward A. Bouchet Graduate Honor Society
2026 Gus W. Weiss Prize for Outstanding Student in Physics, GWU
2024 Parke Prize for Excellence in Theoretical Physics, GWU
2022 Bennhold Family Award for best graduate-student teaching, GWU

Publications

Book

1. **F. Y. Huo**, P. D. Manrique, M. Zheng and N. F. Johnson, *Introduction to Online Complexity: The New Social Physics of Extremes, Misinformation, and AI* (Oxford University Press, 2025).

Journal articles

2. N. F. Johnson and **F. Y. Huo** (equal contribution), Competition for attention predicts good-to-bad tipping in AI, *Patterns*, in press (2026).

3. N. F. Johnson and **F. Y. Huo**, Simple picture of how output from ChatGPT-like AI shifts from good to bad, *PNAS Nexus* **5**, pgag148 (2026).

4. **F. Y. Huo** and N. F. Johnson, Physics of generative AI's atom: Repetition, bias, and beyond, *AIP Advances* **16**, 035305 (2026).

5. **F. Y. Huo**, D. J. Restrepo, P. D. Manrique, G. Woo and N. F. Johnson, Physics reveals and explains patterns in conflict casualties, *EPL* **151**, 12001 (2025).

6. **F. Y. Huo**, P. D. Manrique and N. F. Johnson, Multispecies cohesion: Humans, machinery, AI, and beyond, *Phys. Rev. Lett.* **133**, 247401 (2024).

7. P. D. Manrique, **F. Y. Huo**, S. El Oud and N. F. Johnson, Non-equilibrium physics of multi-species assembly applied to fibrils inhibition in biomolecular condensates and growth of online distrust, *Sci. Rep.* **14**, 21911 (2024).

8. P. D. Manrique, **F. Y. Huo**, S. El Oud, M. Zheng, L. Illari and N. F. Johnson, Shockwavelike behavior across social media, *Phys. Rev. Lett.* **130**, 237401 (2023).

Preprints

9. B. X. Li, **F. Y. Huo** and N. F. Johnson, Interaction creates dynamical AI behavior absent in isolation, arXiv:2608.07457 (2026).

10. N. F. Johnson, **F. Y. Huo** and B. X. Li, Temperature-driven inversion and nonlinear dynamics in ChatGPT-like AIs, arXiv:2608.00939 (2026).

11. **F. Y. Huo** and N. F. Johnson, Many-body tipping dynamics of ChatGPT-like AIs, arXiv:2607.25279 (2026).

12. N. F. Johnson and **F. Y. Huo**, Fusion-fission forecasts when AI will shift to undesirable behavior, arXiv:2605.14218 (2026).

Other publications

13. N. J. Restrepo, D. J. Restrepo, **F. Y. Huo** and N. F. Johnson, Hidden good-to-bad tipping in ChatGPT-like AI, *AI, Computer Science and Robotics Technology* **5**, 1–23 (2026).

14. N. J. Restrepo, D. J. Restrepo, **F. Y. Huo** and N. F. Johnson, When AI output goes bad without anyone noticing: AI tipping and its implications for the legal profession, in *Intelligent Computing: Proceedings of the 2026 Computing Conference*, Vol. 4, Lecture Notes in Networks and Systems **2064**, 161–173 (Springer, 2026).

15. N. J. Restrepo, **F. Y. Huo**, D. J. Restrepo and N. F. Johnson, Going beyond a basic attention head toward an understanding of Transformer-based generative AI, *Adv. Artif. Intell. Mach. Learn.* **5**, 4675–4691 (2025).

16. D. J. Restrepo, **F. Y. Huo**, N. J. Restrepo and N. F. Johnson, Basic attention head as a building block toward understanding Transformer-based generative AI, *Adv. Artif. Intell. Mach. Learn.* **5**, 4518–4531 (2025).

Talks

2026	Online network dynamics ahead of offline shocks: The math of cohesion in human–machine–AI systems. Invited talk, SIAM Annual Meeting (AN26), Cleveland, OH, July 2026.
2025	Physical representation of attention, its properties, and beyond. POLS ENIGMA tutorial seminar, Department of Physics, GWU, April 2025.
2024	The new mathematics of online (mis)behavior. Invited talk, AMS Special Session on Complex Social Systems, Joint Mathematics Meetings, San Francisco, CA, January 2024.

Teaching

2021 – now	**Teaching Assistant, Department of Physics, GWU** Lead group problem-solving in class (SCALE-UP format in the introductory courses); run tutorial-style office hours; grade weekly quizzes and review them with the class. In PHYS 3127, also write the quiz questions.
Fall 2023–26	PHYS 3127, nonlinear dynamics (final-year undergraduate; follows Strogatz)
Spring 2026	PHYS 1022, University Physics II (calculus-based)
Spring 2022	PHYS 1025, University Physics I for life sciences (calculus-based)
Fall 2021	PHYS 1026, University Physics II for life sciences (calculus-based)

Refereeing

Referee for *Physical Review Letters*, *Physical Review E* and *Patterns*.

Media coverage

Patterns paper (2026): *The Independent*, *El País*, *Nautilus*, *Live Science*, *SecurityWeek*, *Decrypt*, and an InvestigateTV report carried by local television stations across the United States.

Earlier work: *APS Physics* (2023); FirstPrinciples.org (2025); *SecurityWeek* (2025).

Computing and languages

Computing: Python (PyTorch, Hugging Face Transformers, NumPy, SciPy) for experiments on open-weight language models
Languages: English, Mandarin Chinese (native)